



2004

TERM 3

PRELIMINARY COURSE EXAMINATION

BIOLOGY

General Instructions • Reading time – 5 minutes • Working time – 120 minutes • Write using black or blue pen • Draw diagrams using pencil • Write your Student Number on the Part A Answer Sheet and the Part B Answerbook.	Total marks for this paper: 70 This paper has two parts, Part A and Part B. Part A Total marks (10) • Attempt all 10 questions in this part • Allow about 20 minutes for this part Part B Total marks (60) • Attempt all questions • Allow about 100 minutes for this part
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PART A

Total Marks (10)

Attempt all questions

1. A characteristic which could be used to distinguish between animals and plants is the presence in all cells of the plant of:
 - a) a cell membrane
 - b) a cell wall
 - c) chloroplasts
 - d) all of the above
2. An apple orchard infected with aphids was sprayed with a new insecticide. The majority of the aphid population was killed but some appeared to be resistant to the insecticide and over time the population increased. This was due to:
 - a) the insecticide causing some aphids to become resistant
 - b) the resistant aphids producing a chemical which inactivated the insecticide
 - c) some aphids having a natural resistance to the insecticide
 - d) those aphids that were killed not being able to mutate quickly enough.
3. Which pair of organisms are most closely related?
 - a) *Pseudomys australis* and *Tadarida australis*
 - b) *Macropus robustus* and *Macropus antilopinus*
 - c) *Phascogale carolinensis* and *Taphozous maculatus*
 - d) *Menura superba* and *Potorous longipes*
4. Stromatolites are found at Shark Bay, Western Australia. These hard, dome-shaped structures are built by which group of organisms?
 - a) Cyanobacteria
 - b) Methanogens
 - c) Protista
 - d) Viruses
5. Which of the following best describes conditions on the early earth about 4.5 billion years ago?
 - a) cold seas, clouds of water vapour, atmosphere of hydrogen, oxygen, carbon dioxide, methane
 - b) warm seas, clouds of water vapour, atmosphere of hydrogen, oxygen, carbon dioxide, methane
 - c) cold seas, clouds of water vapour, atmosphere of hydrogen, ammonia, carbon dioxide, methane
 - d) warm seas, clouds of water vapour, atmosphere of hydrogen, ammonia, carbon dioxide, methane

6. Cyanobacteria belong to a group called:
- eubacteria
 - archaea
 - heterotrophs
 - eukaryotes
7. The earliest life on earth would be approximately:
- 600 million years old
 - 2000 million years old
 - 3800 million years old
 - 5000 million years old
8. Which of the following categories includes the greatest number of different kinds of organisms?
- genus
 - family
 - order
 - class
9. Read the following statements:
- Statement 1: All eukaryotic organisms are multicellular
Statement 2: All multicellular organisms are eukaryotic
Statement 3: All prokaryotic organisms are unicellular
Statement 4: All unicellular organisms are prokaryotic
- The correct statement(s) from the above list is (are):
- 1 only
 - 3 only
 - 2 & 3 only
 - 2 & 4 only
10. Which of the following represent examples of palaeontological and geological evidence for the origin of life on earth?
- macrofossils, proteins and banded iron formations
 - microfossils, macrofossils and red bed rock formations
 - stromatolites, macrofossils and red bed rock formations
 - microfossils, stromatolites and banded iron formations

Student Number.....

Write your student number at the top of this Part B Answer Book

Part B

Total marks (60)

Marks vary for each question

Answer the questions in the space provided

Question 11 (2 marks)

Identify one technological change and describe how this has contributed to increasing our understanding of the origin of life.

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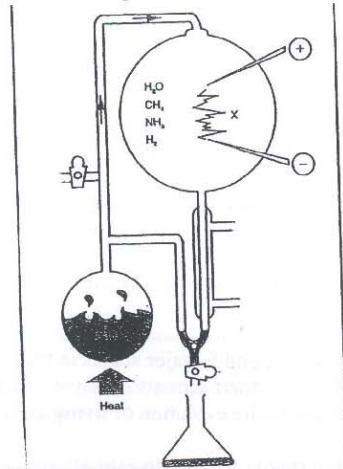
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Question 12 (8 marks)

This diagram represents an experiment first conducted in the 1950's.



a) Who conducted this experiment? (1 mark)

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b) State a reason for this experiment (1 mark)

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c) Describe the experiment (include what X represents) (3 marks)

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d) What was the result of the experiment? (1 mark)

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e) How did this result contribute to hypotheses about the origin of life? (1 mark)

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f) Describe another theory for the origin of the chemicals of life. (1 mark)

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Question 13 (7 marks)

Identify 2 major stages in the origin of life and 5 major stages in the evolution of living things. Record these stages in the order of their formation. Give one example of an organism for each of the 5 major stages in the evolution of living things.

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Question 14 (3 marks)

Explain the influence of the Earth's atmosphere on the evolution of living things.

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Question 15 (6 marks)

- a) Describe the features of the present environment occupied by one of the following: **(2 marks)**

- Archaea
- Eubacteria
- Cyanobacteria
- Nitrogen-fixing bacteria
- Methanogens
- Deep-sea bacteria

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- b) Outline any similarities of the past environment of this organism (from part a) with the present environment **(2 marks)**

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- c) Identify the role of this organism (from part a) in it's ecosystem **(2 marks)**

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Question 16 (7 marks)

In the past, all life was placed into two kingdoms.

- a) What were the two kingdoms? **(1 mark)**

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- b) What criteria was used to classify organisms into these kingdoms? **(1 mark)**

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- c) Outline one disadvantage of a two kingdom classification system **(1 mark)**

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- d) Another classification system is the 5 kingdom system. What are the 5 kingdoms? **(1 mark)**

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- e) Discuss three advantages of the 5 kingdom system **(3 marks)**

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Question 17 (7 marks)

Construct a dichotomous key that could be used to key out the following organisms: mushroom, peaplant, bacterium, fern, starfish, pine tree, moss.

Set out your key in the form 1a, 1b etc

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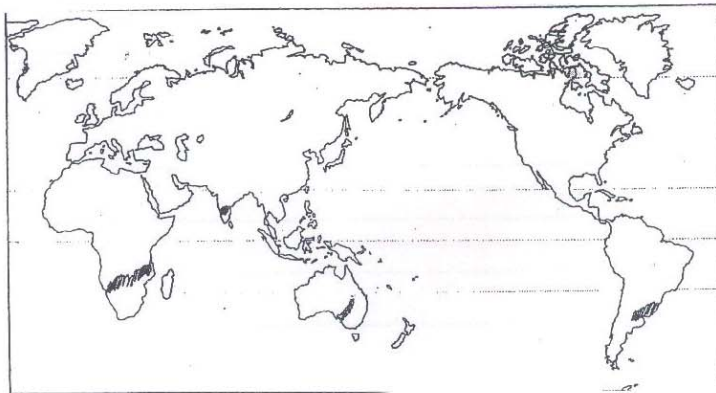
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Question 18 (8 marks)

The map below shows the present day distribution of *Glossopteris* fossils



a) Describe "*Glossopteris*" and its most common type of fossil. (2 marks)

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b) What other continent might you expect to find fossils of *Glossopteris*? (1 mark)

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c) Describe how this fossil could be found on such diverse continents. (3 marks)

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d) Identify two other evidences that support your answer to (c). (2 marks)

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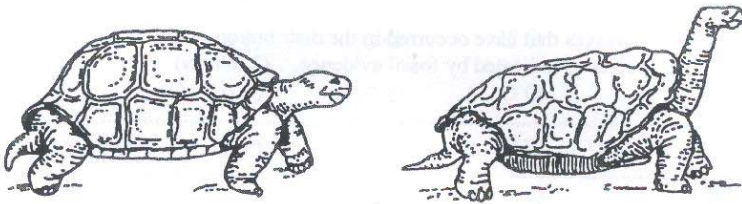
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Question 19 (6 marks)

There are two main types of tortoise on the Galapagos Islands. One has a domed shell and short neck and lives on the moister islands; the other variety has a shell that allows its long neck to be raised.

The long-necked variety lives on the drier islands where the vegetation mainly consists of tall shrubs and bushes.

THE TWO MAIN TYPES OF TORTOISE FOUND ON THE GALAPAGOS ISLANDS



- a) How would Darwin's theory of evolution explain the evolution of the long-necked form of the Galapagos tortoise? **(4 marks)**

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- b) Describe two observations made by Darwin of Australian flora and fauna

(2 marks)

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Question 20 (4 marks)

- a) Identify two changes that have occurred in the distribution of two named Australian species as indicated by fossil evidence. **(2 marks)**

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- b) Discuss two theories to accounts for these changes. **(2 marks)**

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Question 21 (2 marks)

What was the subject of the debate between Huxley and Wilberforce? Outline each of their arguments.

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END OF TEST

Part A Answer Sheet

Write your Student Number at the top of this Part A Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely using ink.

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|-----|------------------------------------|------------------------------------|------------------------------------|--------------------------------------|
| 1. | A ☐ | B ☒ | C ☐ | D ☐ B |
| 2. | A ☐ | B ☐ | C ☒ | D ☐ C |
| 3. | A ☐ | B ☒ | C ☐ | D ☐ <u>B</u> |
| 4. | A ☒ | B ☐ | C ☐ | D ☐ A |
| 5. | A ☐ | B ☐ | C ☐ | D ☒ D |
| 6. | A ☒ | B ☐ | C ☐ | D ☐ <u>A</u> |
| 7. | A ☐ | B ☐ | C ☒ | D ☐ C |
| 8. | A ☐ | B ☐ | C ☐ | D ☒ D |
| 9. | A ☐ | B ☐ | C ☒ | D ☐ <u>C</u> |
| 10. | A ☐ | B ☐ | C ☐ | D ☒ D |

Write your student number at the top of this Part B Answer Book

Part B

Total marks (60)

Marks vary for each question

Answer the questions in the space provided

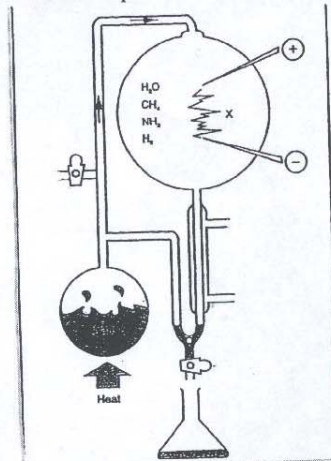
Question 11 (2 marks)

Identify one technological change and describe how this has contributed to increasing our understanding of the origin of life.

- * Radioactive dating - for determining the ages of rocks & fossils
- * deep sea equipment - study variety of deep sea organisms that live at extreme conditions - possible site of the origin of life on Earth
- * Analysis of molecules of DNA, - DNA sequencing, a sequencing. Measurement of these differences could reveal how closely related 2 species may be & reveal how long ago these ancestral lineages went their separate ways.
- * Space travel - extreme conditions under which bacteria can survive - extreme temps, vacuum conditions, lack of water, high radiation levels
- * Microscopes - eg SEM, TEM have assisted in understanding the structure of cyanobacteria
- * computer technology - data bases to hold & manipulate such large amt of info about the genomes of many different species

Question 12 (8 marks)

This diagram represents an experiment first conducted in the 1950's.



a) Who conducted this experiment? (1 mark)

Urey / Miller

b) State a reason for this experiment (1 mark)

to produce organic molecules from simple inorganic molecules in conditions similar to primitive earth's atmosphere.
origin of life!

c) Describe the experiment (include what X represents) (3 marks)

a flask was set up containing the gases (above) to represent the atmosphere of earth early. The gases were exposed to continuous electric discharge (X) representing lightning. The gases were circulated for one week & then the liquid in flask was analysed for evidence for organic molecules.

d) What was the result of the experiment? (1 mark)

production of organic molecules eg amino acids

e) How did this result contribute to hypotheses about the origin of life? (1 mark)

life may have originated on an primitive earth. These organic molecules could eventually form cells.

f) Describe another theory for the origin of the chemicals of life. (1 mark)

Panspermia theory - life originated in outer space & came to earth via meteorites.

Question 13 (7 marks)

Identify 2 major stages in the origin of life and 5 major stages in the evolution of living things. Record these stages in the order of their formation. Give one example of an organism for each of the 5 major stages in the evolution of living things.

1/ organic molecules

2/ membranes

1/ prokaryotic heterotrophic cells eg bacteria

1/ prokaryotic autotrophic cells eg cyanobacteria

1/ eukaryotic cells eg paramecium, amoeba

1/ colonial organisms eg stromatolites, volvox

1/ multicellular organisms eg fish, etc.

Question 14 (3 marks)

Explain the influence of the Earth's atmosphere on the evolution of living things.

- primitive atmosphere - no oxygen - first cells - heterotrophs, used organic molecules as food
- once oxygen accumulated, both heterotrophs & autotrophs could use oxygen for respiration for energy. Eukaryotes increase in diversity of life - inc. in size + complexity of orgs. (then anaerobes more diverse)
- oxygen can also make ozone (protects from harmful effects of UV) - greater diversity of life on land, now protected from UV by ozone

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- Archaea
- Eubacteria
- Cyanobacteria
- Nitrogen-fixing bacteria
- Methanogens
- Deep-sea bacteria

tranquil waters
(stromatolites - shallow sea)
↓
oak bay). ~~perhaps~~

ex. extreme salinity lake, Ca near hot springs.
limited amount of water, & occurrence of CaCO_3

- (2 marks)**

newer stromatolites, oldest known fossil - shallow seas
were more widespread.

mostly aquatic some free living in soil, few types marine
some live in part of life

- (2 marks)**

1) Chloroplast - they produce oxygen - ^{green & ones} responsible for changing them

environment for some to over

- autotrophs! starts the food chain

(also some nitrogen fixing
some in lichens).

(7 marks)

In the past, all life was placed into two kingdoms.

- (1 mark)

plants, animals

- (1 mark)

What criteria was used to classify organisms into these kingdoms? (1 mark)

photosynthetic or heterotrophic Cells w/ or no cell walls
locomotion / no locomotion

- +

do not recognise prokaryotic Vs eukaryotic, difficulty
in classifying fungi, difficulty in classifying microorganisms
with plant & animal features eg. euglenes

- (1 mark)

plants, animals, fungi, protists, monera.

- (3 marks)

my recognises the difference between

Prokaryotes + eukaryotes

- fungi's plants

- recognises no difficulty of classifying unicellular orgs that have plant & animal features eg. euglena.

*may need more space

Question 17 (7 marks)

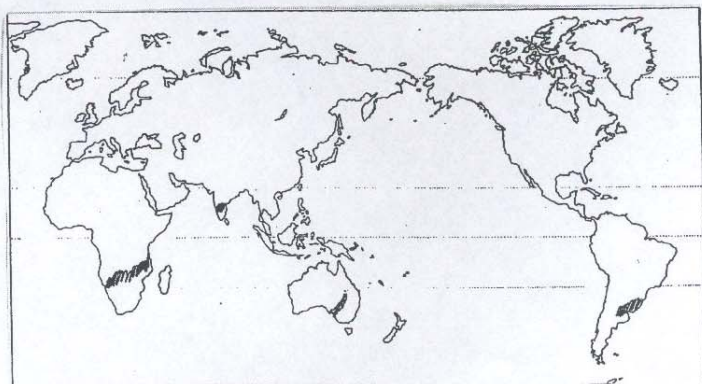
Construct a dichotomous key that could be used to key out the following organisms: mushroom, pea plant, bacterium, fern, starfish, pine tree, moss.

Set out your key in the form 1a, 1b etc

- 1a prokaryotic ----- bacterium
 1b eukaryotic ----- 2
 2a chlorophyll present ----- 3
 2b no chlorophyll ----- 4
 3a produces seeds ----- 5
 3b No seeds ----- 6
 4a Has cones Cell wall present ----- Mushroom
 4b No cones No cell walls ----- Pea plant starfish
 5a No cones ----- pine tree
 5b Cones ----- fern
 6a Xylem present ----- moss
 6b No xylem ----- moss

Question 18 (8 marks)

The map below shows the present day distribution of *Glossopteris* fossils



a) Describe "*Glossopteris*" and its most common type of fossil. (2 marks)

seed fern leaf

b) What other continent might you expect to find fossils of *Glossopteris*? (1 mark)

Antarctica (other than those found on map)

c) Describe how this fossil could be found on such diverse continents. (3 marks)

Evidence for continental drift
 All of these continents formed from same land mass - Gondwana & that these continents were connected during the Permian (300 million yrs ago). *Glossopteris* is an index fossil of the Permian period. The break up occurred around 135 million yrs ago, via the movement of the tectonic plates to present position - separated & yet retaining their fossil *Glossopteris*.

d) Identify two other evidences that support your answer to (c). (2 marks)

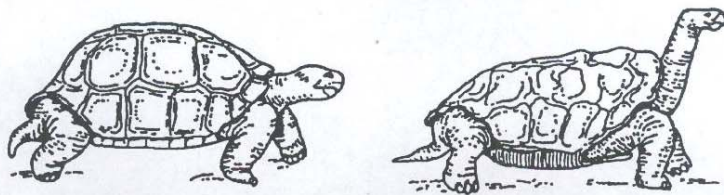
- matching continental margins
- position of mid ocean ridges, spreading zones between cont. plates
- similarities between present day organisms on Gondwana continents (ratites, southern beech tree)

Question 19 (6 marks)

There are two main types of tortoise on the Galapagos Islands. One has a domed shell and short neck and lives on the moister islands; the other variety has a shell that allows its long neck to be raised.

The long-necked variety lives on the drier islands where the vegetation mainly consists of tall shrubs and bushes.

THE TWO MAIN TYPES OF TORTOISE FOUND ON THE GALAPAGOS ISLANDS



- a) How would Darwin's theory of evolution explain the evolution of the long-necked form of the Galapagos tortoise? (4 marks)

* variation
* advantage to get food
* advantageous / disadvantageous traits (short neck)
* survival / reproduced / passed on advantageous traits
* % increased in frequency of long-necked variety on the drier islands

- b) Describe two observations made by Darwin of Australian flora and fauna (2 marks)

* collecting info regarding theory of evolution. $\frac{1}{2}$ for each of 2 each.
flying squirrels from Nth America, platypus.
* Australia's gliders possums
(marsupials) - both native but eat, frog, planarians - egypt.
spherical pores
membranes for flying
deers (marsupials) - kangaroos
(marsupials) - small kangaroos
suffer instead cockatoos etc.
support for "common ancestry".

- a) Identify two changes that have occurred in the distribution of two named Australian species as indicated by fossil evidence. (2 marks)

rainforests have contracted.
subtropical communities & grasslands have spread.

- b) Discuss two theories to account for these changes. (2 marks)

- increasing aridity - less rainforests replaced with subtropical forests
- fire - plants that have fire tolerance - have an advantage - favoured eucalypts rather than rainforest.

Question 21 (2 marks)

What was the subject of the debate between Huxley and Wilberforce? Outline each of their arguments.

Darwin's theory of evolution by natural selection
Wilberforce -

END OF TEST

(196)

Darwin observed similarity between mammals of Australia & other countries. Placentals that live in similar environments & this supported his theory of evolution & common ancestry.
eg. flying squirrels of N. America & glider possum of Australia
eg. deer & kangaroo share similar digestive systems.

21

evolution by
Subject: Darwin's theory of natural selection as applied to human common ancestry with apes.
Wilberforce believed that the idea of man having common ancestry with apes was against the Christian doctrine. He chose to ridicule Huxley by asking whether it was on his father's or mother's side that he claimed to be descended from apes.
Huxley was an advocate of Darwin's theory of evolution. Huxley replied that he would rather be related to an ape than to a man who used his position, eloquence & a few hours acquaintance with biology to ridicule a theory which he did not understand.